
Computational Lithography Solutions for Non-Immersion ArF (193nm) Excimer Laser Deep Ultraviolet Machines for Chemically Amplified Resists with Bottom Anti-Reflective Coating

Executive Program in Semiconductor Manufacturing -
IIT Delhi

Tejas Beedkar
Sushma Shetty
Nitesh Mahawar
Juganta Mosrong

Index

Chapter 1 - Introduction

Chapter 2 - Abbreviations and Concepts

- 2.1 Abbreviations
- 2.2 Diffraction
- 2.3 Chromatic Aberration
- 2.4 Spatial Coherence
- 2.5 Laser Speckle
- 2.6 Full Width at Half Maximum (FWHM)
 - 2.6.1 The Mathematical Framework Behind FWHM
 - 2.6.2 Technical Significance in DUV Lithography

Chapter 3 - ArF Laser Cabinet

- 3.1 ArF Excimer Laser Cabinet
 - 3.1.1 — Beam Conditioning Chain (Raw Excimer Output → Illuminator)

- 3.1.2 Key Beam Characteristics: Source vs. Illuminator-Stage Delivery with Component → Parameter Mapping
- 3.2 Excimer Spectrum (Raw Cavity Output and Ideal)
 - 3.2.1 Practical Excimer Spectrum (Raw Cavity Output)
 - 3.2.2 Ideal Excimer Spectrum for DUV Photolithography
- 3.3 Line Narrowing Module (LNM) / Line Narrowing Package (LNP)
 - 3.3.1 Beam Expander (Prism Pair)
 - 3.3.2 Echelle / Littrow Diffraction Grating
 - 3.3.3 Etalon (Fabry-Pérot)
- 3.4 Metrology - Wavelength Module
- 3.5 Beam Conditioning & Delivery Unit (BDU)
 - 3.5.1 Variable Attenuator
 - 3.5.2 — Beam Expanding Telescope
 - 3.5.3 — Coherence Scrambler / Light Pipe
 - 3.5.4 — DUV Bandpass Filter (KrF systems)

Chapter 4 - Illuminator

- 4.1 Step 1 - Homogenisation
 - 4.1.1 Step 1a - First Fly's Eye Lens Array (Beam Homogenizer #1)
 - 4.1.2 Step 1b - Second Fly's Eye Lens Array (Beam Homogenizer #2)
- 4.2 Step 2 - Pupil Shaping
 - 4.2.1 Step 2a - Diffractive Optical Element (DOE)
 - 4.2.1.1 Function of the DOE
 - 4.2.1.2 DOE matching
 - 4.2.1.3 Diffractive Optical Element (DOE) Types — Pattern Suitability
 - 4.2.2 Step 2b - Partial coherence factor (σ)
 - 4.2.2.1 σ trade-off physics
 - 4.2.2.2 What $\sigma=0$ breaks
 - 4.2.2.3 Outer σ control iris
 - 4.2.3 Mixed Pattern Pupil Shaping
 - 4.2.3.1 Mixed Layer Compensation toolset
 - 4.2.3.2 Evolution by technology node
- 4.3 Step 3 - Relay Lens * 4.2.1 Step 3a - Condenser / Relay Lens Group * 4.2.2 Step 3b - REMA Masking Blades * 4.2.3 Step 3c - REMA relay lens**
- 4.4 Stage 4 - Dose Control

Chapter 5 - Reticle Design in Lithography

- 5.1 Introduction
- 5.2 Reticle Design, Evolution, and AI Integration
 - 5.2.1 Historical Evolution of Reticle Design
 - 5.2.2 Evolution and Mechanics of Modern Reticle Design
 - 5.2.3 Computational Lithography: The Foundation

- 5.2.4 Software: EDA Software and Data Preparation
 - 5.2.5 The Historical and AI Evolution
- 5.3 Advanced Reticle Design and the AI Revolution
 - 5.3.1 The 4x Paradigm and EUV
 - 5.3.2 Types of Reticle Patterns
 - 5.3.3 The AI Revolution
 - 5.3.4 Future Directions: Beyond Monolithic Reticles
- 5.4 The 193nm Epoch: Pushing the Limits of Optical Lithography
 - 5.4.1 Historical Evolution and the 193nm Era
 - 5.4.2 The Rayleigh Criterion
 - 5.4.3 The Shift to Immersion
 - 5.4.4 Resolution Enhancement Techniques (RET)
 - 5.4.5 Economic Context
 - 5.4.6 The 193nm Immersion (193i) Workhorse
- 5.5 Manufacturing-Design Interface
- 5.6 Visualizing Reticle Patterns
- 5.7 Challenges faced by AI-optimized Reticles
- 5.8 Conclusion

Chapter 6: Projection Lens in Reduction Photolithography

- 6.1 Role of the Projection Lens in Reduction Lithography
- 6.2 Computational Lithography and Mask Optimization
- 6.3 Resist Process Co-Optimization and Depth of Focus
- 6.4 Illumination and Recipe Selection Around the Lens
- 6.5 Practical Fine-Tuning of Projection Lenses in Fabs
- 6.6 Primary Failure Modes with Respect to the Projection Lens
 - 6.6.1 Fundamental Resolution / NA Limit
 - 6.6.2 Thermal Distortion and Drift of the Lens
 - 6.6.3 Residual and Evolving Aberrations

Chapter 7 * Wafer and Photoresist Stack

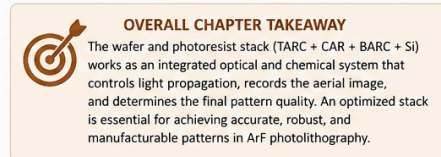
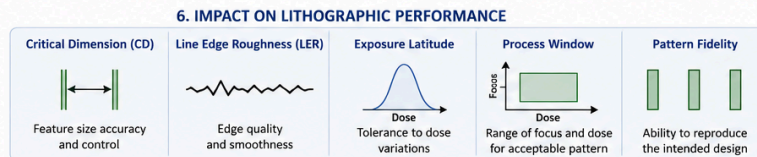
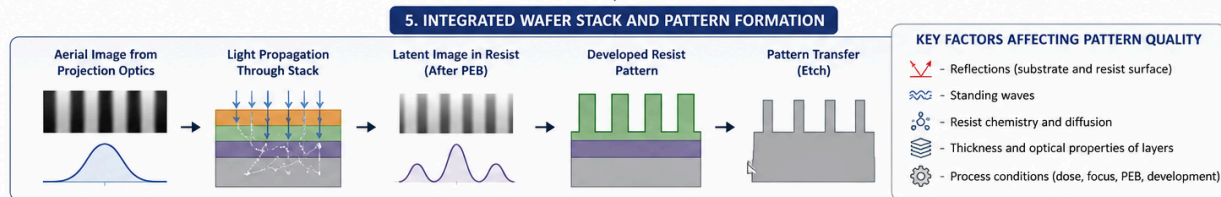
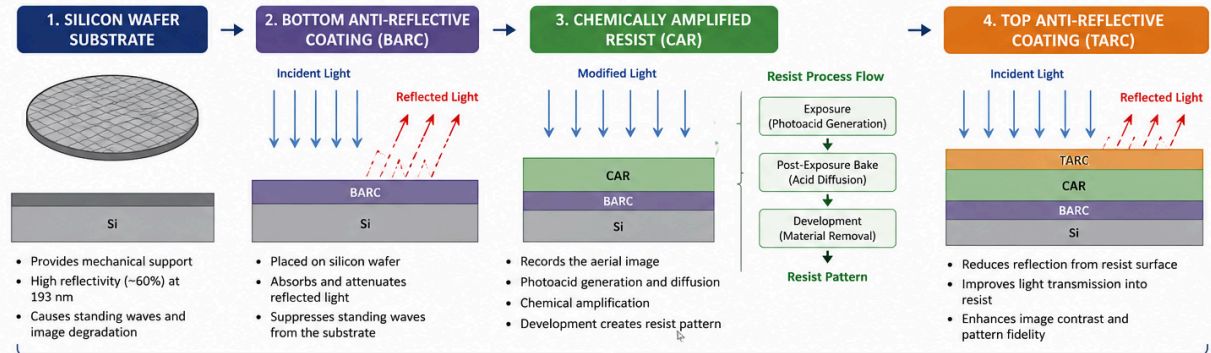
- 7.1 Introduction
- 7.2 Silicon Wafer Substrate
 - 7.2.1 Wafer Foundation
 - 7.2.2 Importance of Wafer Flatness
 - 7.2.3 Optical Interaction with Silicon
- 7.3 Need for Anti-Reflective Coatings
 - 7.3.1 Reflection Mechanism
 - 7.3.2 Standing Wave Formation
 - 7.3.3 Swing Curve Effect

- 7.4 Bottom Anti-Reflective Coating (BARC)
 - 7.4.1 Purpose
 - 7.4.2 Reflection Cancellation Principle
 - 7.4.3 BARC Modeling in Computational Lithography
 - 7.4.4 Impact on Lithographic Performance
- 7.5 Top Anti-Reflective Coating (TARC)
 - 7.5.1 Purpose
 - 7.5.2 Reflection Cancellation Principle
 - 7.5.3 Benefits
 - 7.5.4 TARC Modeling in Computational Lithography
- 7.6 Chemically Amplified Resist (CAR)
 - 7.6.1 Purpose
 - 7.6.2 Resist Composition
 - 7.6.3 Exposure Process
 - 7.6.3.1 Photoacid Generation Model
 - 7.6.4 Latent Image Formation
 - 7.6.4.1 Optical Absorption within the Resist
 - 7.6.5 Post-Exposure Bake (PEB)
 - 7.6.5.1 Acid Diffusion Model
 - 7.6.6 Development
 - 7.6.6.1 Threshold Development Model
 - 7.6.7 CAR Modeling in Computational Lithography
 - 7.6.8 Impact on Lithographic Performance
- 7.7 Integrated Wafer Stack and Pattern Formation
 - 7.7.1 Introduction
 - 7.7.2 Complete Wafer Stack
 - 7.7.3 Pattern Formation Mechanism
 - 7.7.4 Sources of Pattern Degradation
 - 7.7.5 Relationship Between Wafer Stack and Lithographic Performance
- 7.8 Chapter Summary

Chapter 7 Wafer and Photoresist Stack

CHAPTER 9: WAFER AND PHOTORESIST STACK – OVERALL VIEW

Transforming the aerial image into a printed pattern through an optimized wafer stack: BARC + CAR + TARC on a silicon substrate.



7.1 Introduction

Photolithography is the fundamental pattern transfer process used in semiconductor manufacturing. Throughout the lithography system, optical components such as the excimer laser, beam conditioning optics, illuminator, reticle, and projection lens work together to generate a highly controlled aerial image. However, the ultimate objective of these subsystems is not merely the creation of an optical image, but rather the accurate transfer of that image onto a semiconductor wafer where integrated circuit structures are fabricated.

The wafer stage represents the final destination of the optical energy generated by the lithography system. It is at this stage that the aerial image produced by the projection optics is converted into a physical pattern that can subsequently be transferred into underlying device layers through etching and deposition processes. Consequently, the wafer stage serves as the critical interface between optical imaging and semiconductor manufacturing.

In modern Dry ArF (193 nm) photolithography systems, the wafer is not simply a polished silicon substrate. Instead, it consists of a carefully engineered multilayer stack designed to maximize imaging performance and process robustness. A typical wafer stack includes a **Bottom Anti-Reflective Coating (BARC)**, a **Chemically Amplified Resist (CAR)**, and a **Top Anti-Reflective Coating (TARC)**. These layers work together to control optical reflections, improve critical dimension (CD) uniformity, reduce standing-wave effects, and enhance overall pattern fidelity.

The interaction between the projected optical field and the wafer stack is governed by a combination of optical, chemical, and material processes. The aerial image generated by the projection lens initiates photochemical reactions within the resist layer, producing a latent image that is subsequently amplified during post-exposure baking and revealed during development. The final resist pattern therefore depends not only on the quality of the optical image but also on the physical and chemical properties of the wafer materials.

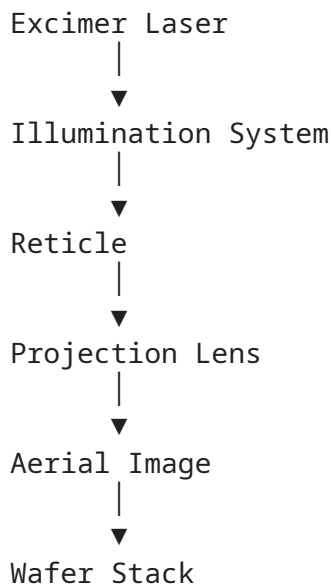
As semiconductor technology continues to scale toward increasingly smaller feature sizes, maintaining accurate focus across the wafer becomes increasingly challenging. Modern lithography scanners therefore incorporate sophisticated focus and leveling systems that continuously monitor wafer topography and dynamically adjust wafer position during exposure. These systems ensure that the wafer surface remains aligned with the focal plane of the projection lens, thereby maximizing image quality and process yield.

From a computational lithography perspective, the wafer stage represents the final simulation target. While preceding modules model illumination, diffraction, mask effects, and optical image formation, the wafer module predicts how the aerial image interacts with the resist stack and ultimately determines the printed pattern. Parameters such as critical dimension (CD), line edge roughness (LER), depth of focus (DOF), and process window are all evaluated at the wafer level to assess lithographic performance.

Therefore, understanding the wafer stack and its associated processes is essential for accurately modeling pattern formation and optimizing the overall lithography workflow. This chapter examines the silicon wafer substrate, anti-reflective coatings, chemically amplified resist systems, focus and leveling mechanisms, pattern formation processes, and the role of wafer modeling in computational lithography.

Chapter Overview

The wafer stage can be viewed as the final pattern formation module within the Dry ArF lithography pipeline:



(BARC / CAR / TARC)



Photoacid Generation



Post Exposure Bake



Development



Final Resist Pattern

The primary goal of the wafer stage is to faithfully convert the optical aerial image into a physical resist pattern while minimizing distortions caused by optical reflections, chemical diffusion, process variations, and focus errors. The effectiveness of this conversion directly determines the quality and manufacturability of the final semiconductor device.

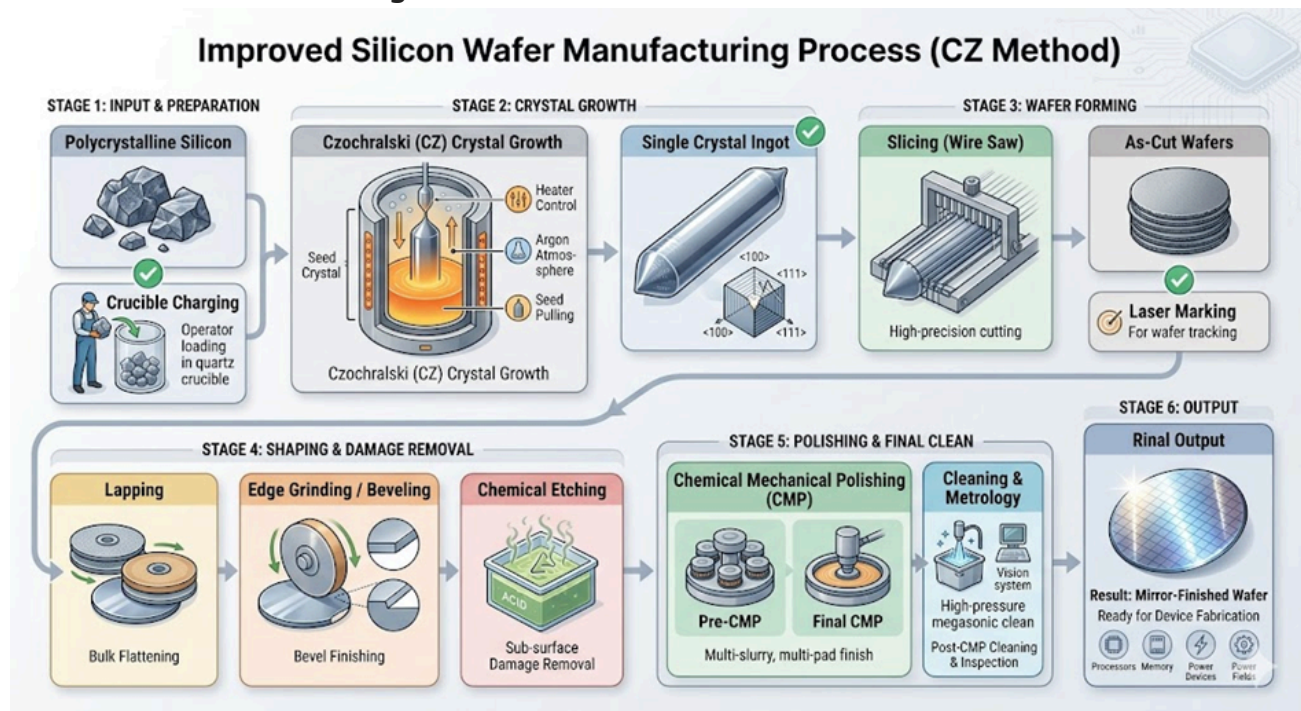
7.2 Silicon Wafer Substrate

7.2.1 Wafer Foundation

The silicon wafer serves as the mechanical and electrical foundation for all integrated circuit structures. Although lithography primarily interacts with the photoresist stack, the underlying wafer strongly influences optical behavior, focus control, thermal stability, and pattern fidelity.

Modern semiconductor manufacturing predominantly uses 300 mm single-crystal silicon wafers fabricated using the Czochralski crystal growth process.

Silicon Wafer Manufacturing Process



The final wafer exhibits exceptional flatness and surface roughness below 1 nm RMS.

7.2.2 Importance of Wafer Flatness

The projection lens can only produce a sharp image when the wafer surface remains within the available depth of focus.

The depth of focus is approximately given by:

$$\text{DOF} \approx k_2 \lambda / \text{NA}^2$$

As numerical aperture increases, depth of focus decreases dramatically.

For ArF lithography operating at ($\lambda = 193$) nm and NA approaching 0.93, the focus budget becomes extremely small. Consequently, wafer flatness becomes a critical parameter.

Sources of focus variation include:

- Wafer bow
- Film stress
- CMP non-uniformity
- Chuck imperfections

Wafer Topography Effects

Ideal Wafer

Lens Focus Plane

Wafer Surface

Non-Ideal Wafer

Lens Focus Plane

Wafer Surface

~~~~~^A^A^A~~~~~

Portions become out of focus

**Key Takeaway:** Wafer flatness is not only a mechanical requirement; it directly determines whether the projected image remains inside the available depth-of-focus budget.

## 7.2.3 Optical Interaction with Silicon

At 193 nm, silicon exhibits significant reflectivity.

Incident Light

↓

Photoresist

↓

Silicon

↑

Reflected Light

These reflections re-enter the photoresist and interfere with incoming light, producing unwanted intensity variations. Therefore, anti-reflective coatings are required to suppress substrate reflections.

## 7.3 Need for Anti-Reflective Coatings

### 7.3.1 Reflection Mechanism

Whenever light encounters a change in refractive index, part of the optical energy is reflected.

Large refractive index differences result in stronger reflections and greater optical distortion.

In lithography, these reflections can occur at:

- Air/TARC interface
- TARC/Resist interface
- Resist/BARC interface
- BARC/Substrate interface

## 7.3.2 Standing Wave Formation

Reflected light interferes with incoming light inside the resist.

Incoming Wave ↓↓↓↓↓↓↓↓↓↓↓↓

Reflected Wave ↑↑↑↑↑↑↑↑↑↑↑↑

Interference Pattern ↓↑↓↑↓↑↓↑↓↑↓↑

This creates standing waves that produce non-uniform exposure through the resist thickness.

Consequences include:

- Sidewall roughness
- Critical dimension variation
- Reduced pattern fidelity

## 7.3.3 Swing Curve Effect

Even when the same mask, exposure dose, and focus settings are used, variations in photoresist thickness can lead to differences in the printed feature size. This phenomenon occurs because reflected light from the substrate interferes with the incoming light inside the resist layer.

As the resist thickness changes, the optical path length traveled by the reflected light also changes. Consequently, the interference conditions within the resist vary from one location to another. Some resist thicknesses produce constructive interference, increasing the effective exposure intensity, while others produce destructive interference, reducing the effective exposure intensity.

Resist Thickness Variations

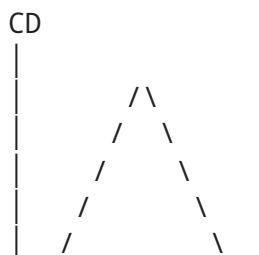
100 nm → CD<sub>1</sub>

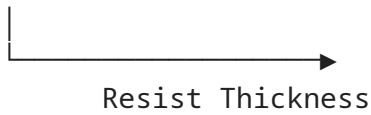
105 nm → CD<sub>2</sub>

110 nm → CD<sub>3</sub>

As a result, identical lithography conditions can generate different critical dimensions (CDs) across the wafer.

The relationship between resist thickness and printed CD is known as the **swing curve**, because the measured CD oscillates as the resist thickness changes.





The swing curve effect is undesirable because small thickness variations can translate directly into linewidth variations, reducing pattern fidelity and process uniformity.

## 7.4 Bottom Anti-Reflective Coating (BARC)

### 7.4.1 Purpose

BARC is positioned between the photoresist and the wafer substrate.

Photoresist

BARC

Silicon Wafer

Its primary function is to absorb reflected light before it re-enters the resist layer.

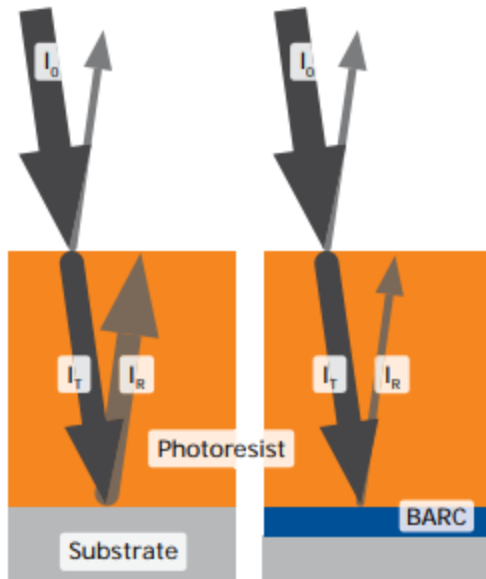
### 7.4.2 Reflection Cancellation Principle

The primary purpose of the Bottom Anti-Reflective Coating (BARC) is to minimize optical reflections originating from the substrate. Without BARC, a portion of the incident exposure light reaches the substrate and is reflected back into the photoresist. These reflected waves interfere with the incoming optical field, creating standing-wave effects, swing curves, and critical dimension (CD) variations.

The reflected light re-enters the photoresist and interferes with the incoming wave, resulting in non-uniform exposure throughout the resist thickness.

To mitigate this effect, a BARC layer is inserted between the photoresist and the substrate.

The BARC material is engineered to possess a suitable refractive index and extinction coefficient so that reflected light is strongly attenuated before it can return to the photoresist.



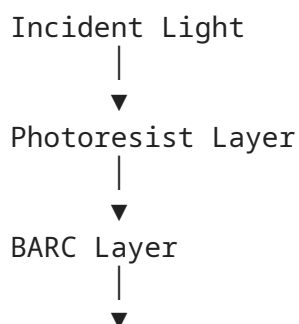
Reflections from the substrate generate a reflected wave ( $I_R$ ) that interferes with the transmitted light ( $I_T$ ) within the photoresist, producing standing-wave effects and exposure non-uniformities. The interference is strongest when the intensities of ( $I_T$ ) and ( $I_R$ ) are similar (left). By introducing a **Bottom Anti-Reflective Coating (BARC)** between the photoresist and substrate (shown in blue, right), the reflected intensity ( $I_R$ ) is significantly reduced. Consequently, the interference between ( $I_T$ ) and ( $I_R$ ) is minimized, resulting in improved exposure uniformity, critical dimension (CD) control, and pattern fidelity.

### 7.4.3 BARC Modeling in Computational Lithography

From a computational lithography perspective, the Bottom Anti-Reflective Coating (BARC) serves as an optical control layer located between the photoresist and the substrate. Its primary function is to suppress substrate reflections that would otherwise re-enter the resist and degrade image quality.

While the projection optics determine the aerial image reaching the wafer, the BARC layer modifies the optical field near the substrate by absorbing and attenuating reflected light. Consequently, BARC modeling plays an important role in predicting standing-wave effects, swing curves, and critical dimension (CD) variations.

The BARC modeling workflow typically consists of the following stages:



Reflection and Absorption



Modified Optical Field



Resist Intensity Distribution

The simulation begins by calculating how light propagates through the resist and BARC layers. Reflections occurring at the substrate interface are partially absorbed by the BARC, reducing the amount of optical energy that returns to the resist.

## Reflection Modeling

Whenever light encounters a change in refractive index, a portion of the optical energy is reflected.

The reflectivity at normal incidence can be approximated using the Fresnel equation:

$$R = \left( \frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

where:

- $R$  = reflected optical power
- $n_1$  = refractive index of the first medium
- $n_2$  = refractive index of the second medium

The objective of the BARC layer is to minimize the effective reflectivity experienced by the resist.

## Optical Absorption within BARC

The BARC material absorbs reflected light as it propagates through the layer.

This absorption can be described using the Beer-Lambert law:

$$I(z) = I_0 e^{-\alpha z}$$

where:

- $I_0$  = incident optical intensity
- $I(z)$  = intensity after traveling distance  $z$
- $\alpha$  = absorption coefficient
- $z$  = propagation distance within the BARC

A larger absorption coefficient results in greater attenuation of reflected light before it can re-enter the photoresist.

## Thickness Optimization

The effectiveness of the BARC layer depends strongly on its thickness.

The optical path length through the film is given by:

$$OPL = n \cdot t$$

where:

- $OPL$  = optical path length
- $n$  = refractive index of the BARC
- $t$  = BARC thickness

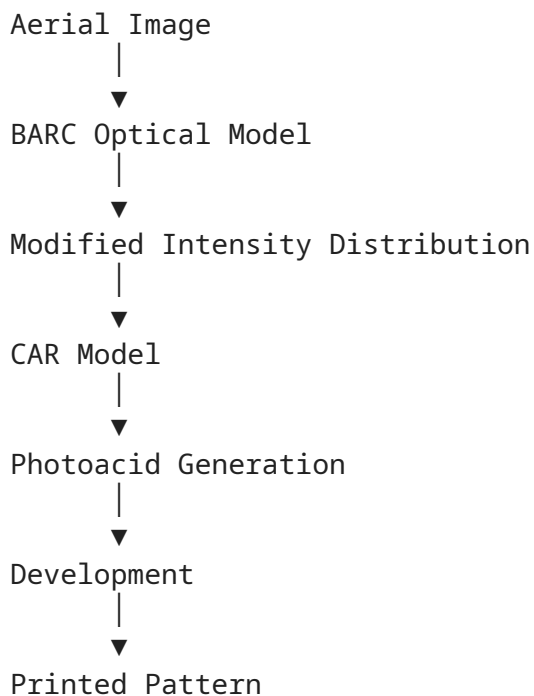
The thickness is selected such that reflected waves are strongly attenuated before reaching the resist layer.

Typical ArF BARC thickness values range from approximately 50 nm to 120 nm depending on the resist stack and substrate materials.

## Computational Lithography Perspective

Within a computational lithography simulator, the BARC model modifies the optical intensity distribution before photoacid generation occurs in the resist.

The complete optical simulation sequence can be represented as:



The simulation evaluates the influence of BARC parameters on:

- Reflectivity
- Standing-wave suppression
- Swing curve magnitude
- Critical Dimension (CD)
- Exposure Latitude
- Process Window

By varying thickness, refractive index, and absorption coefficient, the optimal BARC configuration can be identified before fabrication.

## 7.4.4 Impact on Lithographic Performance

An optimized BARC layer provides several important benefits:

- Reduced substrate reflections
- Lower standing-wave effects
- Reduced swing curve amplitude
- Improved critical dimension control
- Increased exposure latitude
- Larger process window
- Better across-wafer uniformity

As feature sizes continue to shrink, BARC optimization becomes increasingly important because even small optical variations can translate into significant patterning errors.

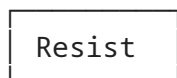
**Key Takeaway:** BARC serves as the optical control layer beneath the photoresist. Through a combination of absorption and thin-film effects, it suppresses substrate reflections, improves image fidelity, and enhances the stability of the lithographic process. Accurate BARC modeling is therefore essential for predicting and optimizing pattern transfer in advanced photolithography systems.

## 7.5 Top Anti-Reflective Coating (TARC)

### 7.5.1 Purpose

A **Top Anti-Reflective Coating (TARC)** is a thin optical layer deposited on top of the photoresist before exposure.

Incident Light



Substrate

During lithography, part of the incident light is reflected at the **air-resist interface** because air and photoresist have different refractive indices. These reflections can create standing waves and intensity variations within the resist, leading to non-uniform exposure.

The primary purpose of TARC is therefore to:

- Reduce reflections at the air–resist interface
- Minimize standing-wave formation
- Improve exposure uniformity throughout the resist thickness
- Enhance critical dimension (CD) control

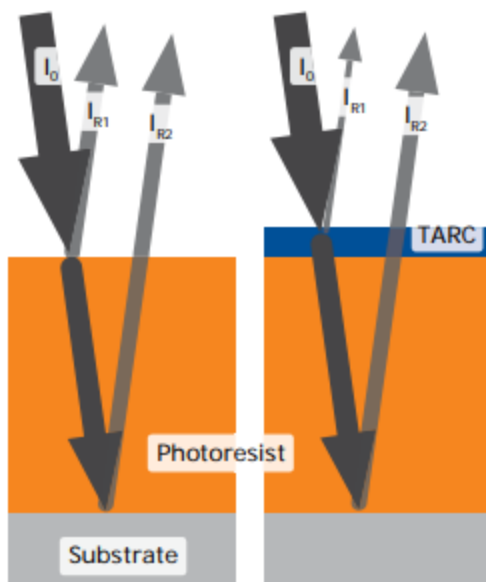
Unlike **BARC**, which suppresses reflections originating from the substrate side, **TARC** suppresses reflections from the top surface of the resist.

## 7.5.2 Reflection Cancellation Principle

TARC functions through the principle of **optical interference**.

When exposure light reaches the TARC layer:

1. A portion is reflected from the air–TARC interface.
2. Another portion travels through the TARC and is reflected from the TARC–resist interface.
3. These reflected waves are designed to have opposite phases.
4. The two reflections destructively interfere, significantly reducing the total reflected intensity.



*The partial beams IR1 and IR2 reflected at the interfaces of air/photoresist and photoresist/substrate can interfere all the more stronger with each other, the more equal their intensities are (left). Right: Through an anti-reflective coating (TARC, shown in blue) on the resist surface, IR1 and thus the interference effect between IR1 and IR2 is minimised*

## 7.5.3 Benefits of TARC

The use of TARC provides several lithographic advantages:

### Improved Critical Dimension (CD) Uniformity



Reduced reflection-induced dose variations produce more consistent feature sizes across the wafer.

### **Enhanced Pattern Fidelity**

Patterns printed in the resist more closely match the intended mask geometry.

### **Reduced Swing Ratio**

Variations in resist thickness have a smaller effect on the printed critical dimension.

### **Increased Exposure Robustness**

The lithography process becomes less sensitive to:

- Focus variations
- Dose fluctuations
- Resist thickness changes

### **Better Line Edge Quality**

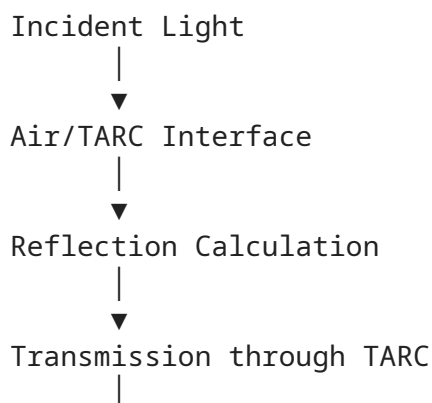
Suppression of standing waves reduces periodic sidewall roughness and improves resist profile control.

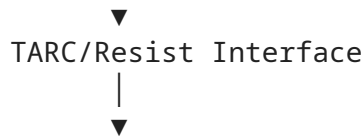
## **7.5.4 TARC Modeling in Computational Lithography**

From a computational lithography perspective, the Top Anti-Reflective Coating (TARC) serves as an optical control layer positioned above the photoresist. Unlike the Chemically Amplified Resist (CAR), which transforms optical energy into a chemical pattern, the TARC modifies the propagation of light before it enters the resist.

The primary objective of TARC modeling is to predict how the coating influences optical reflections, standing-wave effects, and aerial image uniformity. By accurately modeling the TARC layer, lithography engineers can improve critical dimension (CD) control and enhance overall pattern fidelity.

The TARC modeling workflow typically consists of the following stages:





## Resist Intensity Distribution

The simulation begins by calculating reflections occurring at the interfaces between air, TARC, and photoresist. The transmitted optical field is then evaluated to determine the resulting intensity distribution inside the resist layer.

## Reflection Modeling

Whenever light encounters a change in refractive index, a portion of the optical energy is reflected.

The reflectivity at normal incidence can be approximated using the Fresnel equation:

$$R = \left( \frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

where:

- $R$  = reflected optical power
- $n_1$  = refractive index of the first medium
- $n_2$  = refractive index of the second medium

The objective of TARC design is to minimize this reflection.

## Thin-Film Interference Modeling

The TARC layer operates as a thin optical film.

Reflections generated at the:

- Air/TARC interface
- TARC/Resist interface

interfere with one another.

Reflection 1

←←←←←

Reflection 2

→→→→→

Destructive Interference

$\approx 0$  Reflection

By selecting an appropriate thickness and refractive index, these reflections can partially or completely cancel each other.

## TARC Thickness Optimization

The effectiveness of the TARC layer depends strongly on its thickness.

The optical path length through the film is given by:

$$OPL = n \cdot t$$

where:

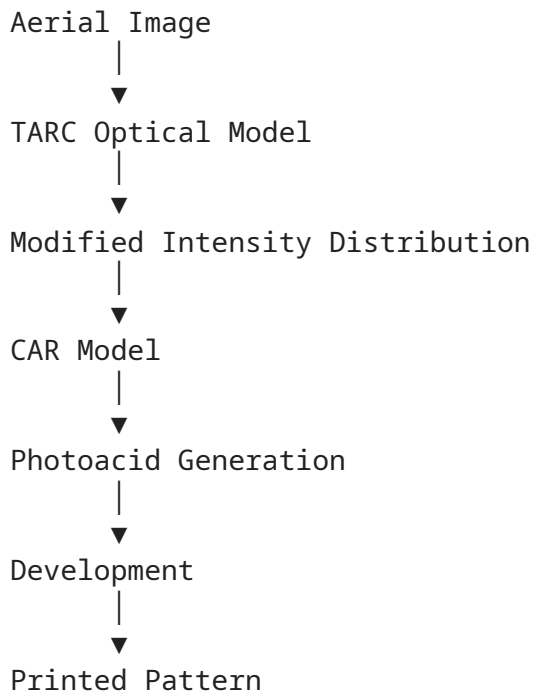
- $OPL$  = optical path length
- $n$  = refractive index of the TARC
- $t$  = TARC thickness

The thickness is optimized such that reflected waves undergo destructive interference, minimizing the amount of light reflected back toward the illumination system.

### Computational Lithography Perspective

Within a computational lithography simulator, the TARC model modifies the optical intensity distribution before photoacid generation occurs in the resist.

The complete optical-resist simulation sequence can be represented as:



Thus, the TARC model acts as a pre-processing stage for the resist simulation.

### Impact on Lithographic Performance

The outputs of TARC modeling are used to evaluate:

- Reflectivity
- Standing-wave suppression
- Swing curve reduction
- Critical Dimension (CD) uniformity

- Exposure latitude
- Process window

An optimized TARC layer produces a more uniform optical intensity distribution within the resist, resulting in improved pattern fidelity and greater process robustness.

### Key Takeaway

**Top Anti-Reflective Coatings (TARCs)** reduce reflections originating from the **air-resist interface** through destructive optical interference. By minimizing top-surface reflections, TARC improves CD control, pattern fidelity, swing ratio, and process stability. Together with **BARC**, which suppresses substrate reflections, TARC helps ensure that the photoresist receives a uniform and predictable optical dose from both the top and bottom interfaces of the lithographic stack.

## 7.6 Chemically Amplified Resist (CAR)

### 7.6.1 Purpose

The Chemically Amplified Resist (CAR) is the photosensitive layer responsible for recording the optical image projected onto the wafer. It serves as the primary imaging medium within the lithographic stack and ultimately determines how accurately the aerial image is translated into a physical resist pattern.

Unlike conventional photoresists, CAR systems utilize chemical amplification mechanisms that significantly increase sensitivity to incident radiation. This enables high-throughput manufacturing while maintaining the resolution required for advanced semiconductor devices.

The primary functions of the CAR layer are:

- Capture the aerial image generated by the projection optics.
- Convert optical intensity into chemical information.
- Amplify the photochemical reaction through acid catalysis.
- Generate a latent image that can be developed into a physical pattern.
- Provide sufficient etch resistance for subsequent process steps.

As feature sizes continue to shrink, the performance of the resist increasingly becomes a limiting factor in lithographic pattern fidelity.

### 7.6.2 Resist Composition

A typical chemically amplified resist consists of several carefully engineered components that work together to transform optical energy into a printable pattern.

Chemically Amplified Resist

└─ Polymer Matrix

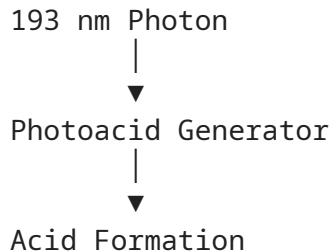
- Photoacid Generator (PAG)
- Quencher
- Solvent

### **Polymer Matrix**

The polymer matrix forms the structural backbone of the resist. It provides mechanical stability and determines the dissolution behavior during development.

### **Photoacid Generator (PAG)**

The PAG is responsible for generating acid molecules when exposed to 193 nm radiation.



The generated acid initiates subsequent chemical reactions within the resist.

### **Quencher**

Quenchers regulate acid diffusion and prevent unwanted chemical reactions in unexposed regions. They improve pattern fidelity and reduce image blur.

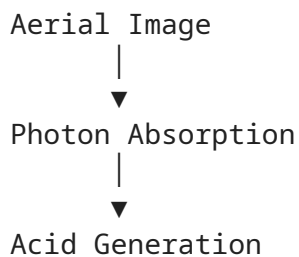
### **Solvent**

The solvent enables uniform resist coating during spin coating and is largely removed during the soft-bake process.

## **7.6.3 Exposure Process**

During exposure, the aerial image produced by the projection lens enters the resist layer.

Regions receiving higher optical intensity generate larger concentrations of acid molecules through the PAG mechanism.



### **Photoacid Generation Model**

The concentration of generated acid is proportional to the optical dose absorbed by the resist:

$$H(x, y) = \alpha I(x, y)t$$

where:

- $H(x, y)$  = generated acid concentration
- $\alpha$  = resist sensitivity constant
- $I(x, y)$  = aerial image intensity
- $t$  = exposure time

This equation establishes the connection between the aerial image produced by the optical system and the chemical reactions occurring within the resist.

Higher optical intensity results in greater acid generation and stronger chemical modification during subsequent processing steps.

## 7.6.4 Latent Image Formation

Immediately after exposure, no visible pattern exists within the resist.

Instead, a chemical image known as the latent image is formed.

Optical Image



Acid Distribution



Latent Image

The latent image stores the information required to create the final resist pattern.

### Optical Absorption within the Resist

The optical intensity decreases as light propagates through the resist thickness according to the Beer-Lambert Law:

$$I(z) = I_0 e^{-\alpha z}$$

where:

- $I_0$  = incident optical intensity
- $I(z)$  = intensity at depth  $z$
- $\alpha$  = absorption coefficient
- $z$  = depth within the resist

Consequently, acid generation is not perfectly uniform throughout the resist thickness. The resulting three-dimensional acid concentration distribution contributes to the formation of the latent image.

## 7.6.5 Post-Exposure Bake (PEB)

The post-exposure bake activates the chemical amplification mechanism.

During PEB, acid molecules diffuse through the resist and catalyze reactions that modify the polymer structure.

Acid



Catalytic Reaction



Polymer Modification

One acid molecule can trigger many chemical reactions, greatly increasing resist sensitivity.

## Acid Diffusion Model

During Post-Exposure Bake (PEB), generated acid molecules diffuse through the resist and catalyze polymer modification reactions.

The diffusion process is commonly described using Fick's Second Law:

$$\frac{\partial H}{\partial t} = D \nabla^2 H$$

where:

- $H$  = acid concentration
- $D$  = diffusion coefficient
- $t$  = bake time
- $\nabla^2 H$  = spatial concentration gradient

The diffusion coefficient determines how far acid molecules spread during baking.

A larger diffusion coefficient produces smoother features but may also blur small patterns. Conversely, insufficient diffusion can increase line edge roughness.

## 7.6.6 Development

Following Post-Exposure Bake, the resist is immersed in a developer solution.

In a positive-tone resist, exposed regions become more soluble and dissolve during development.

Before Development



After Development



## Threshold Development Model

A simplified development model assumes that regions with acid concentrations above a threshold dissolve during development:

$$M(x, y) = \begin{cases} 1, & H(x, y) > H_{th} \\ 0, & H(x, y) \leq H_{th} \end{cases}$$

where:

- $M(x, y)$  = developed resist profile
- $H(x, y)$  = acid concentration
- $H_{th}$  = dissolution threshold

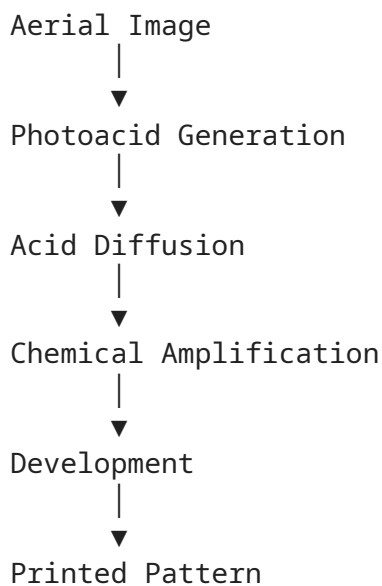
This threshold operation converts the continuous chemical latent image into a discrete physical resist structure.

The resulting pattern serves as the etch mask for subsequent semiconductor processing steps.

### 7.6.7 CAR Modeling in Computational Lithography

From a computational lithography perspective, the Chemically Amplified Resist (CAR) serves as the bridge between the optical image produced by the projection system and the final printed resist pattern. While the optical subsystem determines the aerial image intensity distribution reaching the wafer, the resist model predicts how this optical information is transformed into a physical structure through a sequence of chemical and diffusion processes.

The resist modeling workflow typically consists of the following stages:



The simulation begins with the aerial image intensity distribution, which determines the amount of acid generated within the resist. During the post-exposure bake, acid molecules diffuse through the resist and catalyze polymer modification reactions. The resulting chemical distribution is subsequently processed through a development model to predict the final resist profile.



Mathematically, the resist modeling process can be represented as:

$$I(x, y) \rightarrow H(x, y) \rightarrow M(x, y) \rightarrow CD$$

where:

- $I(x, y)$  represents the aerial image intensity.
- $H(x, y)$  represents the generated acid concentration.
- $M(x, y)$  represents the developed resist profile.
- $CD$  represents the final printed critical dimension.

The primary objective of CAR modeling is to predict how process parameters influence the final printed pattern. Typical simulation outputs include:

- Critical Dimension (CD)
- Line Edge Roughness (LER)
- Resist Profile Shape
- Process Window
- Exposure Latitude
- Depth of Focus (DOF)

By incorporating resist behavior into computational lithography simulations, engineers can evaluate pattern fidelity and process robustness before fabrication. This significantly reduces experimental development time and enables optimization of lithographic performance through virtual process exploration.

**Key takeaway:** CAR modeling transforms the optical aerial image into a predicted resist pattern by simulating photoacid generation, diffusion, chemical amplification, and development. It therefore forms one of the most important components of modern computational lithography workflows.

## 7.7 Integrated Wafer Stack and Pattern Formation

### 7.7.1 Introduction

The individual layers discussed in the previous sections do not operate independently during photolithography. Instead, the Bottom Anti-Reflective Coating (BARC), Chemically Amplified Resist (CAR), and Top Anti-Reflective Coating (TARC) function together as an integrated wafer stack. Their combined purpose is to transform the aerial image generated by the projection lens into a high-fidelity resist pattern that accurately reproduces the reticle geometry.

While BARC and TARC primarily perform optical functions by suppressing unwanted reflections, the CAR layer performs the chemical conversion of optical energy into a printable pattern. The

overall lithographic performance therefore depends on the interaction of all three layers.

### 7.7.2 Complete Wafer Stack

A typical Dry ArF wafer stack consists of the following layers:

Air



TARC



CAR



BARC



Silicon Substrate

**Functions of Individual Layers**

| Layer   | Primary Function                           |
|---------|--------------------------------------------|
| TARC    | Reduces reflections at the resist surface  |
| CAR     | Converts optical image into chemical image |
| BARC    | Suppresses substrate reflections           |
| Silicon | Mechanical and electrical substrate        |

### 7.7.3 Pattern Formation Mechanism

The primary objective of the wafer stack is to transform the aerial image generated by the projection optics into a physical resist pattern that accurately reproduces the intended mask geometry. This transformation occurs through a sequence of optical, chemical, and material processes within the TARC, CAR, and BARC layers.

The complete pattern formation sequence can be represented as:

Aerial Image



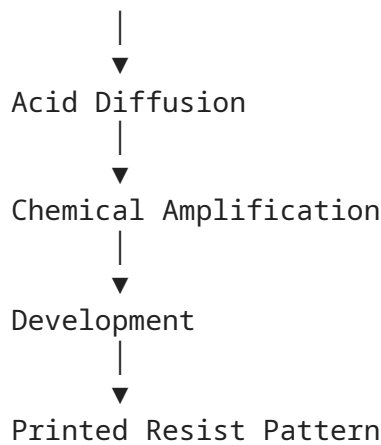
Optical Propagation through TARC/CAR/BARC



Photon Absorption



Photoacid Generation

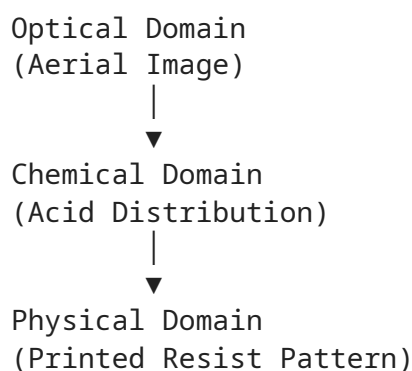


During exposure, the aerial image enters the wafer stack through the TARC layer. The TARC minimizes reflections at the resist surface, allowing a more uniform optical field to enter the photoresist. The optical energy then propagates through the CAR layer, where photons are absorbed by Photoacid Generator (PAG) molecules. Simultaneously, the BARC layer suppresses substrate reflections, preventing reflected light from interfering with the incoming optical field.

The absorbed photons generate acid molecules within the resist. The resulting acid concentration distribution corresponds to the optical intensity distribution of the aerial image and forms a latent chemical image. During the Post-Exposure Bake (PEB), acid molecules diffuse through the resist and catalyze chemical reactions that modify the polymer structure. This chemical amplification mechanism significantly increases resist sensitivity and enables high-resolution patterning.

Following PEB, the wafer undergoes development. In positive-tone chemically amplified resists, the exposed regions become more soluble and are selectively removed by the developer solution. The latent chemical image is thereby converted into a physical resist structure.

The overall process can be viewed as a transformation of information across three domains:



The quality of the final resist pattern depends on the combined performance of the TARC, CAR, and BARC layers. Any optical distortions, reflection effects, diffusion variations, or development non-uniformities can alter the final feature dimensions and reduce pattern fidelity.

## 7.7.4 Sources of Pattern Degradation

Several mechanisms can degrade pattern quality during this process:

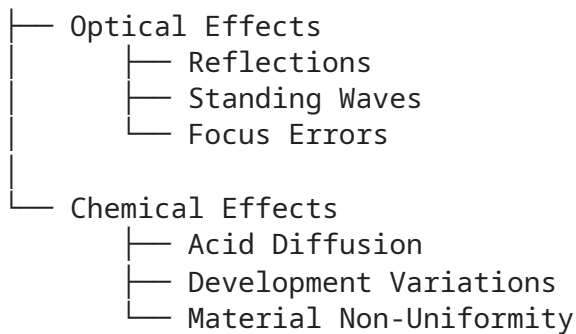
## Optical Effects

- Substrate reflections
- Standing waves
- Swing curves
- Focus variations

## Chemical Effects

- Excessive acid diffusion
- Non-uniform development
- Stochastic effects
- Resist thickness variation

## Pattern Errors



The purpose of the BARC, TARC, and CAR stack is to minimize these error sources and improve pattern fidelity.

## 7.7.5 Relationship Between Wafer Stack and Lithographic Performance

The primary objective of the wafer stack is not merely to record the optical image generated by the projection optics, but to ensure that the final printed pattern accurately reproduces the intended reticle geometry. The quality of the printed pattern is therefore determined by the combined behavior of all layers within the wafer stack.

TARC



Reflection Control

CAR



Image Recording

BARC



Reflection Suppression

Combined Effect

↓

## Pattern Fidelity

Each layer performs a specific function, yet the final lithographic result depends on their collective interaction.

The TARC layer minimizes reflections occurring at the top surface of the resist, thereby improving the uniformity of the optical field entering the photoresist. The BARC layer suppresses substrate reflections that would otherwise generate standing-wave effects and swing curve variations. Positioned between these two optical control layers, the CAR records the aerial image and converts it into a physical resist structure through photoacid generation, diffusion, and development.

Consequently, the wafer stack directly influences several key lithographic performance metrics.

## Critical Dimension (CD)

Critical Dimension (CD) refers to the width of a printed feature and is one of the most important parameters in semiconductor manufacturing.

Target CD

↓

40 nm

Printed CD

↓

39.8 nm

Variations in optical intensity, substrate reflections, or resist chemistry can cause deviations between the intended and printed dimensions. By suppressing reflections and improving image uniformity, the wafer stack helps maintain consistent CD values across the wafer.

## Line Edge Roughness (LER)

Line Edge Roughness (LER) describes deviations along the edges of printed features.

Ideal Edge

Actual Edge

~~~~~^~~~~~

LER is strongly influenced by resist chemistry, acid diffusion, and development processes. Excessive roughness can degrade transistor performance and increase manufacturing variability. The CAR layer therefore plays a critical role in determining edge quality.

Pattern Fidelity

Pattern fidelity measures how closely the final resist pattern matches the original mask design.

Mask Pattern

↓

Printed Pattern

Any distortion caused by optical reflections, diffusion effects, or development non-uniformities reduces pattern fidelity. The combined action of TARC, CAR, and BARC is therefore essential for preserving the integrity of the original design.

Exposure Latitude

Exposure latitude describes the range of exposure doses that produce acceptable feature dimensions.

Nominal Dose

±

Acceptable Variation

A well-designed wafer stack reduces sensitivity to small dose fluctuations, allowing the lithography process to operate more robustly under manufacturing conditions.

Process Window

The process window represents the range of focus and exposure conditions under which acceptable patterns can be reliably produced.

Focus Variation

+

Dose Variation

+

Material Variation

↓

Process Window

An optimized wafer stack enlarges the process window by reducing the influence of optical reflections and resist-related variations. A larger process window generally leads to higher manufacturing yield and improved process stability.

Connection to Computational Lithography

The ultimate objective of computational lithography is to predict the final printed pattern before fabrication takes place. To achieve this, simulations combine optical modeling of the wafer stack with resist modeling of the chemical processes occurring within the photoresist.

The overall simulation workflow can be represented as:

Wafer Stack
(TARC + CAR + BARC)

↓

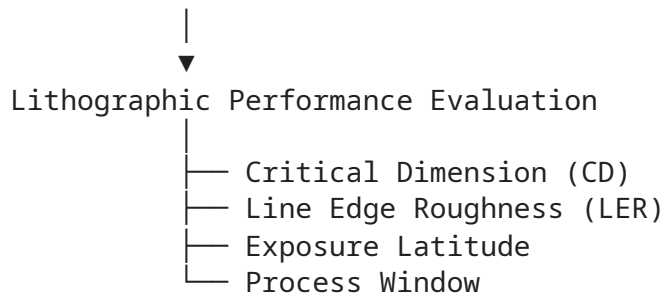
Optical Modeling

↓

Resist Modeling

↓

Printed Pattern Prediction



The optical model calculates how light propagates through the wafer stack, taking into account reflection suppression by the TARC and BARC layers as well as optical absorption within the photoresist. The resulting optical intensity distribution serves as the input to the resist model.

The resist model then simulates photoacid generation, acid diffusion, chemical amplification, and development processes within the CAR layer to predict the final resist profile.

Once the printed pattern has been predicted, various lithographic performance metrics can be extracted. These metrics provide quantitative measures of process quality and manufacturing robustness. For example, Critical Dimension (CD) evaluates feature size accuracy, Line Edge Roughness (LER) characterizes edge smoothness, Exposure Latitude measures tolerance to dose variations, and the Process Window defines the range of operating conditions that produce acceptable patterns.

Consequently, computational lithography does not merely simulate optical intensity distributions. Its primary purpose is to predict how the wafer stack influences final pattern quality and manufacturing performance before fabrication is performed.

Key Takeaway The wafer stack functions as an integrated optical and chemical system. TARC controls reflections at the resist surface, BARC suppresses reflections from the substrate, and CAR converts the optical image into a physical pattern. Together, these layers enable accurate pattern transfer and form the foundation of modern computational lithography simulations.

7.8 Summary

This chapter examined the wafer stage of the Dry ArF photolithography system, which serves as the critical interface between optical image formation and semiconductor pattern fabrication. Unlike the upstream lithography components that generate and project the aerial image, the wafer stack is responsible for converting this optical information into a physical resist pattern that can subsequently be transferred into device layers through etching and other fabrication processes.

The chapter began with an overview of the silicon wafer substrate and its influence on lithographic performance. The importance of wafer flatness, optical interactions with silicon, and the challenges associated with substrate reflections were discussed. These reflections were shown to be a major source of standing-wave effects, swing curves, and critical dimension variations, motivating the need for anti-reflective coatings.

The roles of the Bottom Anti-Reflective Coating (BARC) and Top Anti-Reflective Coating (TARC) were then examined. BARC was shown to suppress reflections originating from the substrate through optical absorption and thin-film effects, while TARC reduces reflections at the resist surface through reflection cancellation mechanisms. Together, these coatings improve optical uniformity within the resist and enhance overall pattern fidelity.

The Chemically Amplified Resist (CAR) was subsequently analyzed as the primary imaging layer of the wafer stack. The processes of photoacid generation, latent image formation, acid diffusion, chemical amplification, and development were presented to illustrate how the aerial image is transformed into a physical resist structure. Mathematical models describing optical absorption, acid generation, and diffusion were introduced to establish the connection between lithographic imaging and computational modeling.

The chapter concluded by integrating the functions of BARC, TARC, and CAR into a unified wafer stack model. The complete pattern formation sequence—from optical propagation through the stack to the final developed resist profile—was discussed, together with the factors that influence lithographic performance. Particular emphasis was placed on critical dimension control, line edge roughness, exposure latitude, process window, and pattern fidelity, as these metrics ultimately determine the manufacturability of semiconductor devices.

Overall Chapter Takeaway: The wafer stack functions as an integrated optical and chemical system in which BARC, TARC, and CAR work together to transform the aerial image generated by the projection optics into a manufacturable semiconductor pattern. The effectiveness of this transformation directly determines lithographic performance and serves as the foundation for computational lithography analysis and process optimization.